

# Major Project Documentation

## Week 1 – Field Visit and Initial Discussion Report

**Institution:** Maharaja Institute of Technology Mysore

**Date:** 2 March 2026

**Location:** AIISH (All India Institute of Speech and Hearing), Mysore

### Purpose of the Visit

The primary objective of the visit to **AIISH (All India Institute of Speech and Hearing)** was to investigate the practical hurdles encountered by people with speech and language impairments. By engaging directly with clinical experts, the team sought to bridge the gap between theoretical knowledge and the lived realities of non-verbal patients. Furthermore, the initiative focused on identifying opportunities for **assistive technology** to enhance communicative autonomy and patient outcomes.

### Discussion Summary

#### Option 1: Structured Clinical Summary

**Objective:** Addressing Expressive Communication Barriers

- **The Core Challenge:** Patients suffering from strokes or speech impairments often retain full linguistic comprehension while losing the ability to speak. This disconnect between understanding and expression is a primary focus of treatment.
- **Therapeutic Approaches:** While **telepractice** offers convenience, the doctor emphasized that **in-person speech therapy** remains the gold standard. Direct interaction allows clinicians to monitor subtle behavioral cues and non-verbal feedback that are often lost over video calls.
- **The Power of Non-Verbal Cues:** In the absence of speech, therapists rely heavily on a patient's "silent language"—specifically gestures, eye contact, and facial expressions—to gauge intent and progress.

### Gaps Identified in the Current System

The most significant hurdle is the **comprehension-expression mismatch**. Patients, particularly those recovering from strokes, often maintain cognitive awareness and language understanding but lack the motor or neurological ability to verbalize thoughts. Bridging this "silent gap" remains a complex clinical challenge.

## Communication Gap

A significant portion of the non-verbal population remains "silenced" not by their condition, but by the **scarcity of affordable and user-friendly communication aids**. Without accessible assistive technology, these individuals find it nearly impossible to convey even basic needs or complex thoughts to their caregivers and peers

## Therapy Accessibility Gap

Traditional speech therapy imposes a heavy **logistical burden** on patients, as it necessitates regular travel to clinical facilities. While **tele-rehabilitation** has emerged as a convenient alternative, it often fails to replicate the clinical depth and efficacy of face-to-face intervention.

## Limited Assistive Technologies

Although sophisticated solutions like eye-tracking hardware are available, their **prohibitive costs** create a significant divide. Consequently, these life-changing tools remains out of reach for a vast majority of patients, particularly those in underserved communities.

## Limited AI Integration

Current communication systems rarely utilize **Artificial Intelligence** to bridge the gap for non-verbal patients. There is a profound need for AI-driven platforms that can autonomously decode **facial micro-expressions, ocular patterns, and physical gestures**, converting these silent signals into actionable communication in real-time.

## Project Status

While a definitive **problem statement** for the major project remains under development, the visit has provided a fertile ground for exploration. The team is currently synthesizing the insights gathered to evaluate various domains within **assistive communication technology**. We intend to finalize our research trajectory through a combination of literature review, data analysis, and

ongoing dialogue with clinical specialists.

## **Conclusion**

The immersion at **AIISH** was instrumental in uncovering the multifaceted hurdles within the speech-language pathology landscape. By pinpointing significant deficiencies in current assistive frameworks, the visit underscored a clear opportunity for **technological intervention**.

Specifically, the potential for **Artificial Intelligence** to revolutionize communication support was a key takeaway, serving as the cornerstone for our upcoming research and development efforts.